Deploy and Attach EBS on Windows EC2:

Prerequisites

- An AWS account
- A running **Windows EC2 instance**
- IAM permissions to manage EC2 and EBS

Steps to Deploy and Attach EBS on Windows EC2

1. Login to AWS Console

- Go to https://console.aws.amazon.com
- Navigate to **EC2 Dashboard**

2. Create an EBS Volume

1. Go to **Volumes** (under EC2 → Elastic Block Store)
2. Click **"Create Volume"**
3. Configure:
   - **Size** (e.g., 10 GiB)
   - **Volume type** (e.g., gp2 or gp3 for general purpose)
   - **Availability Zone** → **Must match** the zone of your Windows EC2 instance (e.g., `ap-south-1a`)
4. Click **Create Volume**

3. Attach EBS Volume to Windows EC2

1. After volume is created, **select it**
2. Click **"Actions"** → **Attach Volume**
3. Choose:
   - **Instance**: Select your Windows EC2
   - **Device name**: Leave as default (e.g., `sbd` or `sdg`)
4. Click **Attach**

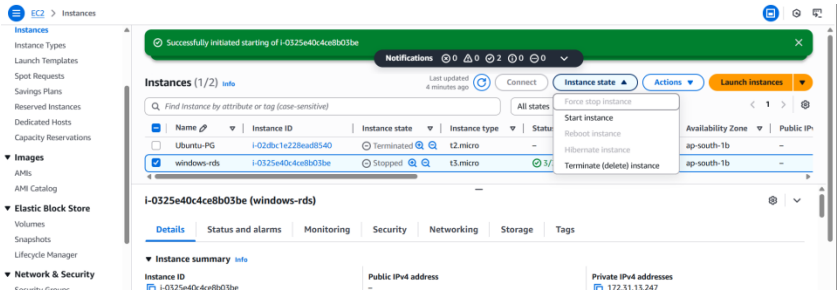
4. Login to Windows EC2 Instance

- Use **Remote Desktop (RDP)** to connect
- Go to **"This PC"** → **Manage** → **Disk Management**

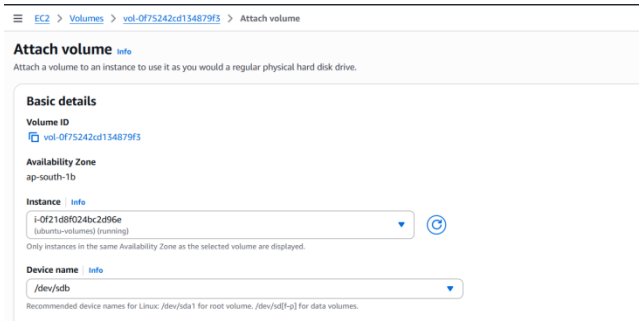
5. Initialize & Format the Volume

1. It will show a **new uninitialized disk**
2. Right-click → **Initialize Disk**
   - Use **GPT** for larger disks, else **MBR**
3. Create a **New Simple Volume**
   - Assign drive letter (e.g., `D:`)
   - Choose **NTFS**
   - Format and finish

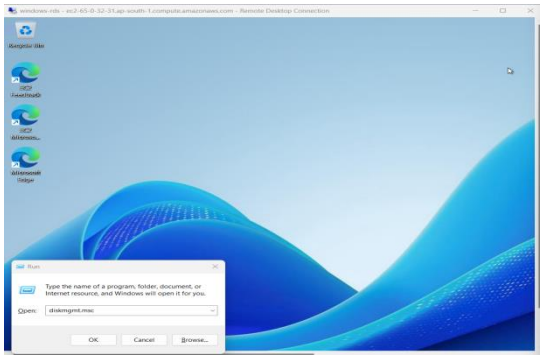
Now the EBS volume is usable on your Windows EC2 instance!



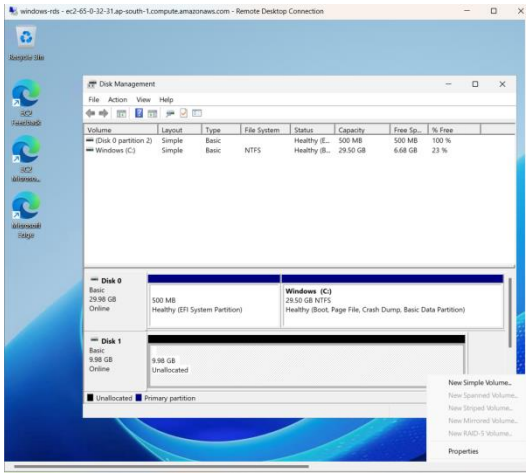
INSTANCE CREATION



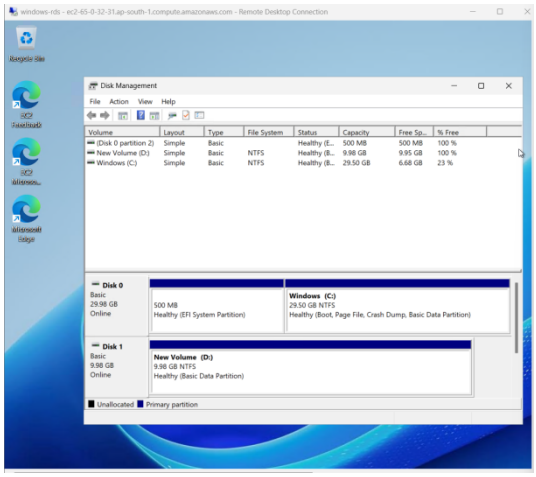
VOLUME ATTACHIING



DISK MGMT



CONFIGURATION



D DRIVE SETUP

Deploy and Attach EBS on Linux EC2:

Prerequisites

- A **running Linux EC2 instance**
- IAM permissions to manage EC2 and EBS
- Key pair (.pem) file to SSH into your instance

Steps to Deploy and Attach EBS on Linux EC2

1. Create EBS Volume

1. Go to **EC2 → Volumes**
2. Click **"Create Volume"**
3. Choose:
   - **Size** (e.g., 8 GiB)
   - **Type**: gp2/gp3
   - **Availability Zone**: Must match your EC2 (e.g., `ap-south-1a`)
4. Click **Create Volume**

2. Attach Volume to Linux EC2

1. Select the volume
2. Click **Actions → Attach Volume**
3. Select your Linux EC2 instance
4. Device name: e.g., `/dev/xvdf`
5. Click **Attach**

3. SSH into Your Linux EC2

```
ssh -i your-key.pem ec2-user@<Public-IP>
```

4. Verify EBS Attachment

```
lsblk
```

Check if the volume is already formatted

bash
CopyEdit
sudo file -s /dev/xvdd

- Output will show data if it's unformatted.
- If it already shows ext4 or another filesystem, **do not format** again (you'll lose data).

5. Format the Volume (only if it's unformatted)

sudo mkfs -t ext4 /dev/xvdd
This formats the EBS volume with the **ext4** file system.

6. Create a Mount Point Directory

sudo mkdir /mnt/data_volume
This is the folder where you'll mount the volume.

7. Mount the Volume

sudo mount /dev/xvdd /mnt/data_volume
Now you can use /mnt/data_volume as a regular folder with the new EBS space.

---

8. Verify the Mount

df -h
You should see /dev/xvdd mounted on /mnt/data_volume with its size and usage.

✅(Optional but Recommended) Make the Mount Persistent
a. Get the UUID:

sudo blkid /dev/xvdd

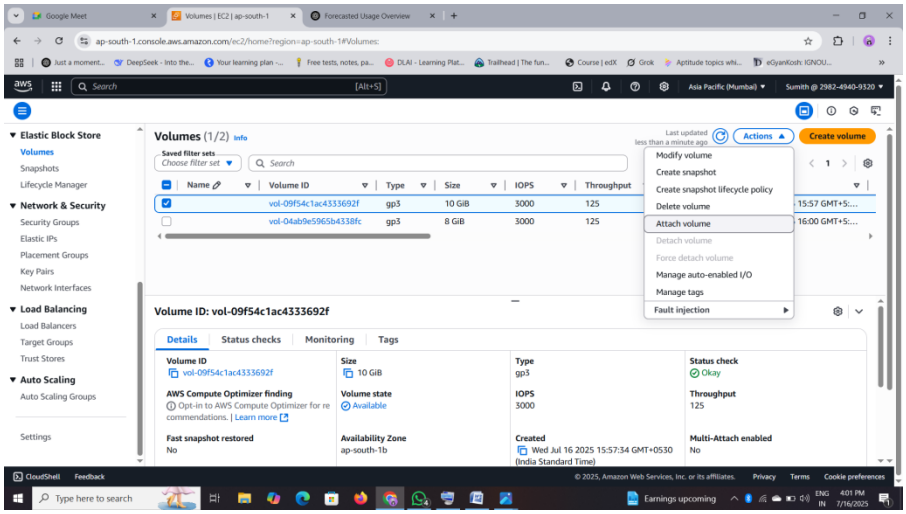
b. Add entry to /etc/fstab:

sudo nano /etc/fstab
Add this line (replace <UUID> with actual value):

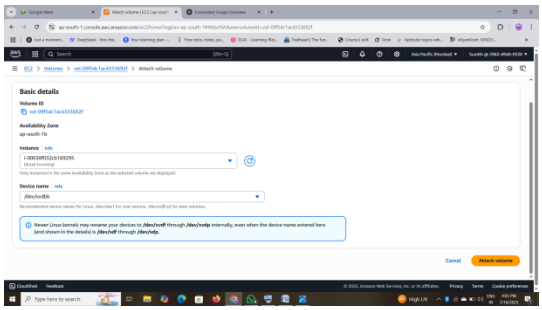
UUID=<your-uuid> /mnt/data_volume ext4 defaults,nofail 0 2

c. Test it:

sudo mount -a
If no error → successful setup.



VOLUME ATTACHING


VOLUME SETUP

```
ubuntu@ip-172-31-2-232:~$ history
    1  sudo apt update
    2  lsblk
    3  sudo file -s /dev/xvdd
    4  sudo mkfs -t ext4 /dev/xvdd
    5  sudo mkdir /mnt/data_volume
    6  sudo mount /dev/xvdd /mnt/data_volume
    7  df -h
    8  history
ubuntu@ip-172-31-2-232:~$
```

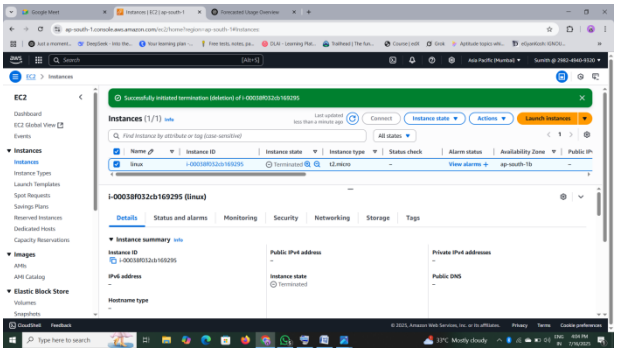
COMMANDS

```
Last login: Wed Jul 16 05:38:57 2025 from 183.82.110.216
ubuntu@ip-172-31-2-232:~$ lsblk
NAME     MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0      7:0    0 27.2M  1 loop /snap/amazon-ssm-agent/11320
loop1      7:1    0 73.9M  1 loop /snap/core22/1981
loop2      7:2    0 50.9M  1 loop /snap/snapd/24505
xvda     202:0    0    8G  0 disk
├─xvda1  202:1    0    7G  0 part /
├─xvda14 202:14   0    4M  0 part
├─xvda15 202:15   0  106M  0 part /boot/efi
└─xvda16 259:0    0  913M  0 part /boot
xvdd     202:48   0   10G  0 disk
ubuntu@ip-172-31-2-232:~$ sudo file -s /dev/xvdd
/dev/xvdd: data
ubuntu@ip-172-31-2-232:~$ sudo mkfs -t ext4 /dev/xvdd
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: ee4fa72c-0c18-452d-98de-6da64c0cc2fe
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

ubuntu@ip-172-31-2-232:~$ sudo mkdir /mnt/data_volume
ubuntu@ip-172-31-2-232:~$ sudo mount /dev/xvdd /mnt/data_volume
ubuntu@ip-172-31-2-232:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       6.8G  2.0G  4.8G  30% /
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  868K  191M   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
/dev/xvda16     881M   86M  734M  11% /boot
/dev/xvda15     105M  6.2M   99M   6% /boot/efi
tmpfs            96M   12K   96M   1% /run/user/1000
/dev/xvdd       9.8G   24K  9.3G   1% /mnt/data_volume
ubuntu@ip-172-31-2-232:~$
```

OUTPUT OF D DRIVE



TERMINATING INSTANCES